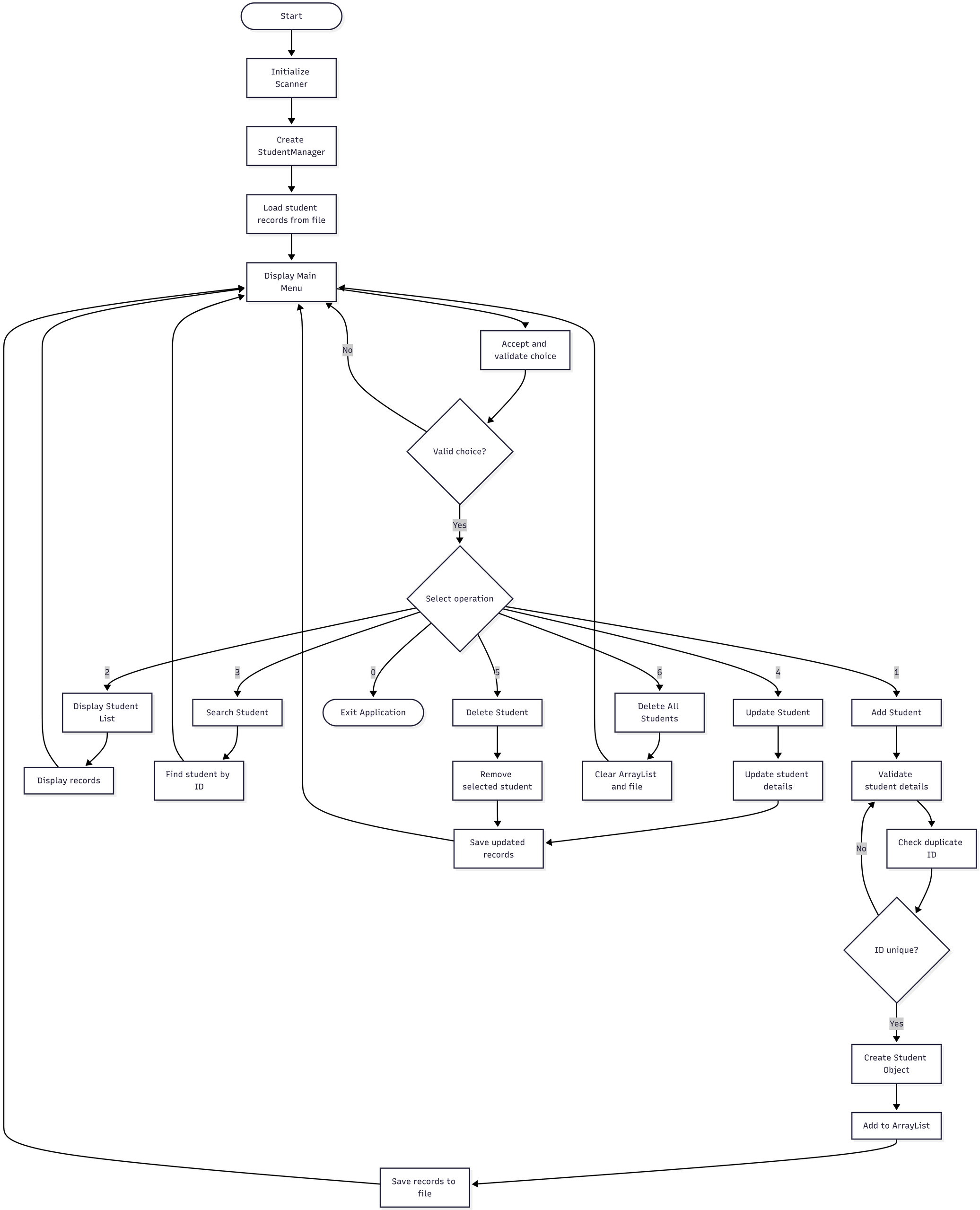

Start
Initialize Scanner
Create StudentManager
Load student records from file
Display Main Menu
Accept and validate choice
No
Valid choice?
Yes
Select operation
2
Display Student List
Display records
3
Search Student
Find student by ID
0
Exit Application
5
Delete Student
Remove selected student
6
Delete All Students
Clear ArrayList and file
4
Update Student
Update student details
1
Add Student
Validate student details
No
Check duplicate ID
ID unique?
Yes
Create Student Object
Add to ArrayList
Save updated records
Save records to file